Q1. Distance Class

Problem

A surveyor needs a program to add and compare distances measured in **feet** and **inches**.
Create a class named **Distance** with the following:

Private Data Members

- `feet` (int)
- `inches` (float)

Methods

- `setData(int feet, float inches)` – assigns values.
- `display()` – displays the distance in the format: `5 feet 8.5 inches`
- `addDistance(Distance d)` – returns a new `Distance` object representing the sum of two distances.
- `compareDistance(Distance d)` – compares two distances and displays whether the current object is **greater than**, **less than**, or **equal to** the other distance.

Example

```
Distance 1 = 5 feet 8.5 inches
Distance 2 = 3 feet 10.5 inches

Total Distance = 9 feet 7.0 inches
Distance 1 is greater than Distance 2.
```

Hint

- Convert every **12 inches into 1 foot** while adding.
- Compare total inches (feet × 12 + inches).

Q2. Student Result Processing

Problem

A school wants to calculate the percentage and GPA of students.
Create a class named **Student** with the following data members:

- Name
- Class
- Roll Number
- Marks in 5 subjects

Methods

- Calculate total marks
- Calculate percentage
- Calculate GPA (use the grading system given by your teacher)
- Display student details

Create **5 Student objects** and display their results.

Example

```
Name : Ram
Roll : 12

Marks
English : 78
```

```
Math : 85
Science : 91
Computer : 80
Social : 76

Total : 410
Percentage : 82%
GPA : 3.6
```

Hint

Store subject marks in an array.

Q3. Invoice System

Use Case

A hardware store wants software to prepare customer invoices.
Create a class named **Invoice**.

Data Members

- Part Number
- Part Description
- Quantity
- Price per Item

Requirements

- Provide getter and setter methods.
- Implement '`getInvoiceAmount()`' which returns '`Quantity × Price`'

Create another class **InvoiceTest** to test the program.

Example

```
Part Number : A101
Description : Keyboard
Quantity : 4
Price : 1500

Invoice Amount = 6000
```

Hint

If quantity or price is negative, store it as **0**.

Q4. Employee Salary System

Use Case

A company wants software to calculate yearly salaries and annual increments.
Create an **Employee** class.

Data Members

- First Name
- Last Name

- Monthly Salary

Requirements

- Getters and setters
- Calculate yearly salary
- Increase salary by **10%**
- Display yearly salary before and after the raise.

Create **EmployeeTest** class.

Example

```
Employee : Hari Sharma

Monthly Salary = 40,000

Yearly Salary = 480,000

After 10% Raise

Monthly Salary = 44,000

Yearly Salary = 528,000
```

Hint

```
Yearly Salary = Monthly Salary × 12
```

Q5. Bank Account Management System

Use Case

A bank wants a simple application to manage customer accounts.
Create a class named **BankAccount**.

Data Members

- Depositor Name
- Account Number
- Account Type (Saving/Current)
- Account Balance

Methods

- Initialize account information
- Deposit money
- Withdraw money (only if sufficient balance exists)
- Display account details
- Display current balance

Example

```
Account Holder : Sita

Balance : Rs. 25,000

Deposit : Rs. 5,000

Current Balance : Rs. 30,000
```

```
Withdraw : Rs. 10,000

Remaining Balance : Rs. 20,000
```

Hint

Before withdrawal:

```
If withdrawal amount > balance
Display: "Insufficient Balance"
```

Otherwise,

```
balance = balance - withdrawal
```